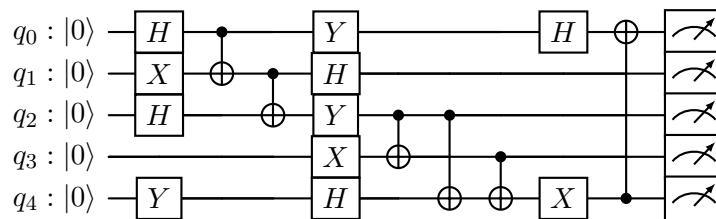


Worksheet: 5-Qubit Circuit Implementation and Measurement

Instructions. Build the exact circuit shown below on 5 qubits, measure all qubits, and run the circuit for at least 1024 shots on a simulator.



Gate order summary

1. Apply H on q_0 , X on q_1 , H on q_2 , and Y on q_4 .
2. Apply CNOT with control q_0 and target q_1 .
3. Apply CNOT with control q_1 and target q_2 .
4. Apply Y on q_0 , H on q_1 , Y on q_2 , X on q_3 , and H on q_4 .
5. Apply CNOT with control q_2 and target q_3 .
6. Apply CNOT with control q_2 and target q_4 .
7. Apply CNOT with control q_3 and target q_4 .
8. Apply H on q_0 and X on q_4 .
9. Apply CNOT with control q_4 and target q_0 .
10. Measure all 5 qubits.

Tasks

1. Draw the circuit in Qiskit and verify that it matches the worksheet diagram.
2. Run the circuit for 1024 shots and write down the measurement counts.
3. List the three most frequent output bitstrings.
4. Does the output distribution look concentrated on a few bitstrings or spread out across many bitstrings?
5. Remove the final CNOT (control q_4 , target q_0), run again, and compare the counts with the original circuit.

Reflection questions

1. Which gates in this circuit mainly create superposition?
2. Which gates are responsible for creating correlations between qubits?
3. Which single-gate modification do you think would change the output distribution the most? Why?